

Species Status Assessment

Common Name: Brook Trout

Date Updated: September 2024

Scientific Name: *Salvelinus fontinalis*

Updated by: Eli Lagacy/Nick Jordan

Class: Actinopterygii

Family: Salmonide

Species Synopsis (a short paragraph which describes species taxonomy, distribution, recent trends, and habitat in New York):

Brook Trout is a species of charr in the class Actinopterygii and the family Salmonidae. Brook Trout are native to cold water streams and lakes in northeastern North America. Brook Trout have been introduced throughout the world for angling opportunities and exist on almost every continent (MacCrimmon & Campbell 1969). Many wild populations currently only occupy headwater streams and clean secluded freshwater lakes. Less commonly, they can exhibit an anadromous life history, with populations undergoing migrations from salt or brackish waters into freshwaters to spawn (Snook et al. 2015). Water quality is the key limiting factor for Brook Trout, which require cold, clean, well-oxygenated water (Smith 1985; Stauffer et al. 2016). They also require clean gravel substrate, free of silt for spawning (Stauffer et al. 2016). Brook Trout are sensitive to changes in land use associated with deforestation, agriculture, and urbanization (Kanno et al. 2015). Disruptions of forested riparian areas, habitat degradation and fragmentation, increased summer water temperature and introduction of non-native fish species have resulted in reduction of habitat availability. Habitat degradation and disconnected aquatic connectivity caused by impassable road stream crossings have largely confined Brook Trout to headwaters with limited human disturbance. In the Eastern United States Brook Trout have been extirpated from 28% of native subwatersheds with populations decreasing by more than half in an additional 35% of subwatersheds (Hudy et al. 2008). Restoration and enhancement efforts by Federal and State agencies, in collaboration with non-profit organizations work to conserve self-sustaining populations in the Brook Trout's native home range. Wild Brook Trout populations in New York have been relatively consistent in recent decades with a moderate increase in detections from the mid-2000's through early 2010's, and a decrease in detections over the last decade returning to levels of the 1990's and early 2000's. However, this may be indicative of sampling effort rather than a large-scale population increase.

I. Status

a. Current legal protected Status

i. **Federal:** None **Candidate:** No

ii. **New York:** SGCN

b. Natural Heritage Program

i. **Global:** G5 (Secure)

ii. **New York:** S5 (Secure) **Tracked by NYNHP?:** Yes

Other Ranks:

-IUCN Red List: N/A

Status Discussion:

In Connecticut Brook Trout are not a state listed species but are categorized as an SGCN. Here, Brook Trout are primarily found in headwater streams. The Connecticut Department of Energy and Environmental Protection (CTDEEP) released a trout management plan for Connecticut's Rivers and Streams (Hyatt et al. 1999). This document distinguishes the difference between wild

populations and stocked fish. Additionally, the document calls for stream restoration and enhancement to increase habitat suitability for wild fish. In Connecticut, wild trout management areas and trout management areas are heavily monitored. Monitoring efforts in Connecticut compare historic wild populations of Brook Trout with current presence-absence surveys (Eltz & Beauchene 2020), and stocking programs currently exist to supplement fishing opportunity for anglers. Based on survey data between 1988-1994 and 2018-2019, Brook Trout were present at only 68/107 sites in 2018-2019 as compared to historic data in 1988-1994. Additionally, several sites they were still present at exhibited declines in density as well (Eltz & Beauchene 2020). The authors stated that more sites are needed to be assessed into the future to make more confident determinations of trends, but since these sites were randomly selected there is a likelihood that populations have actually declined in the state within this timeframe. NatureServe lists Brook Trout as “Secure” in Connecticut (NatureServe 2024).

In Massachusetts Brook Trout are not a state listed species but are categorized as an SGCN. Naturally reproducing (wild) Brook Trout are widespread in the western and central portions of the state, but have experienced declines in occurrence and abundance in the eastern portions primarily due to urbanization and development. Sea run “Coasters” historically found in Massachusetts Bay have been extirpated, but small numbers of sea run Brook Trout still remain and are heavily managed in tributaries of Buzzards Bay (Snook et al. 2015). Stocked Brook Trout can be found throughout much of the state. NatureServe lists Brook Trout as “Apparently Secure” in Massachusetts (NatureServe 2024).

In New Jersey Brook Trout are not a state listed species but are categorized as an SGCN. According to the New Jersey Coldwater Fisheries Management Plan (NJCFMP; 2005), there are wild reproducing trout populations in 121 streams in the northern half of the state, in 94 of these Brook Trout are the only trout species present. The distribution of Brook Trout would potentially be more widespread if not for the introduction of Brown Trout to New Jersey, which has pushed some Brook Trout populations into smaller headwater reaches (NJCFMP 2005). In a 2004 study completed by the Eastern Brook Trout Joint Venture (EBTJV), it was determined that New Jersey had 79 subwatersheds that contained Brook Trout, and that they were extirpated from 97 (55% of the original range in New Jersey; NJCFMP 2005). In 2018, the New Jersey Fish and Wildlife Department adopted regulations that all Brook Trout in the Brook Trout Conservation Zone (which encompasses the majority of their native range in the state) should be released unharmed when caught (New Jersey DEP 2024). NatureServe lists Brook Trout as “Vulnerable” in New Jersey (NatureServe 2024).

In Pennsylvania Brook Trout are not a state listed species but are categorized as an SGCN. According to the EBTJV in 2006, Pennsylvania had the most robust Brook Trout population in the mid-Atlantic (Stauffer et al. 2016). While populations have been extirpated in many historical subwatersheds in the state, robust populations in the northern and central regions still persist (Stauffer et al. 2016). Between 2020-2024 the Pennsylvania Fish and Boat Commission (PFBC) have plans to formally assess the density and occurrence of wild Brook Trout in the state as compared to historical data (PFBC Trout Management Plan 2020). Additionally, where practical, the PFBC will manage cold water streams for wild Brook Trout, and hatchery raised Brook Trout will be reduced statewide to allow for wild strain Brook Trout to persist (PFBC Trout Management Plan 2020). NatureServe lists Brook Trout as “Apparently Secure” in Pennsylvania (NatureServe 2024).

In Vermont Brook Trout are not a state listed species but are categorized as an SGCN. A statewide evaluation of wild Brook Trout published in 2017 found that their present-day condition supports abundant natural reproduction with multiple age-classes present and healthy (Kirn 2017). The Vermont Management Plan for Brook, Brown, and Rainbow Trout (2016) prioritizes wild trout

populations where possible. NatureServe lists Brook Trout as “Secure” in Vermont (NatureServe 2024).

In Ontario Brook Trout are present in the majority of the province, with the densest populations in the southwest portions north and northeast of Lake Huron (Wood 2017; Holm et al. 2021). They are also present in the far western portions of the province where they are non-native (Holm et al. 2021). Additionally, there are sea-run populations of Brook Trout in tributaries of the Hudson Bay in the far northern portions of the province (Ontario Ministry of Natural Resources 2014). NatureServe lists Brook Trout as “Secure” in Ontario (NatureServe 2024).

In Quebec Brook Trout are presumed to be doing well, although limited information exists on Brook Trout populations in this province. While a large province with a multitude of regions and levels of anthropogenic alteration, Quebec contains some populations of Brook Trout that are regarded as “pristine” (Fraser & Bernatchez 2008). This is most likely due to most of the province being incredibly remote and unaffected by direct alterations from human development. NatureServe lists Brook Trout as “Apparently Secure” in Quebec (NatureServe 2024).

II. Abundance and Distribution Trends

a. North America

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 10 Years

b. Northeastern U.S. (USWFS Region 5)

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 10 Years

c. Adjacent States and Provinces

CONNECTICUT

Not Present: _____

No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: _____ Unknown: ☒

ii. Distribution

Declining: _____ Increasing: _____ Stable: _____ Unknown: ☒

Time Frame Considered: 35 Years

Listing Status: None – S5 (Secure) SGCN?: Yes

MASSACHUSETTS

Not Present: _____

No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____
Time Frame Considered: 10 Years
Listing Status: None – S4 (Apparently Secure) SGCN?: Yes

NEW JERSEY Not Present: _____ No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: _____ Unknown: ☒

ii. Distribution

Declining: _____ Increasing: _____ Stable: _____ Unknown: ☒

Time Frame Considered: 10 Years

Listing Status: None – S3 (Vulnerable) SGCN?: Yes

PENNSYLVANIA Not Present: _____ No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 10 Years

Listing Status: None – S4 (Apparently Secure) SGCN?: Yes

VERMONT Not Present: _____ No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 10 Years

Listing Status: None – S5 (Secure) SGCN?: Yes

ONTARIO Not Present: _____ No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 10 Years

Listing Status: None – S5 (Secure) SGCN?: N/A

QUEBEC Not Present: _____ No Data: _____

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: N/A

Listing Status: N/A SGCN?: N/A

d. New York

i. Abundance

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

ii. Distribution

Declining: _____ Increasing: _____ Stable: ☒ Unknown: _____

Time Frame Considered: 30 Years

Monitoring in New York (*specify any monitoring activities or regular surveys that are conducted in New York*):

Regular surveys are performed by fisheries staff pertaining to economically valuable cold-water fishes, such as Brook Trout. Catch rate orientated trout stocking (CROTS) surveys have been completed regularly around the state since 1990. While the goals of this survey type are to gain better information on the capacity of a stream reach to support trout generally, they also yield valuable information on Brook Trout abundance and distribution. Additionally, monitoring programs are carried out as needed by the NYSDEC Rare Fish Unit.

Trends Discussion (*insert map of North American/regional*):

Brook Trout populations in North America are relatively stable as much of their native range is in Canada and the northeastern portions of the United States, where their populations are generally doing well (Stauffer et al. 2016, Holm et al. 2021; NatureServe 2024). However, much of the knowledge about their population status in some portions of their range (particularly large swaths of Canada) are based on limited empirical data across the landscape (Haxton et al. 2019). Ideally, more research should be done spatially to confirm their “stable” population status here.

Brook Trout have established themselves as an introduced non-native species across much of the mountain west from New Mexico to Alaska, and even in several other continents (U.S. Forest Service 2007; Alaska Department of Fish & Game 2008; Kershner et al. 2019; Fish & Game New Zealand 2024). However, within portions of their southern native range, their populations have become extirpated or significantly reduced, and remaining populations occur in small, isolated populations making them prone to more local extirpations (Kazyak et al. 2022; NatureServe 2024).

In New York while Brook Trout trends are reduced as compared to historical data, over the past several decades their populations and distributions have been relatively stable. A moderate increase in detections occurred from 2004 – 2013, however rather than a large-scale population increase and subsequent decrease, this may be a construct of sampling effort.

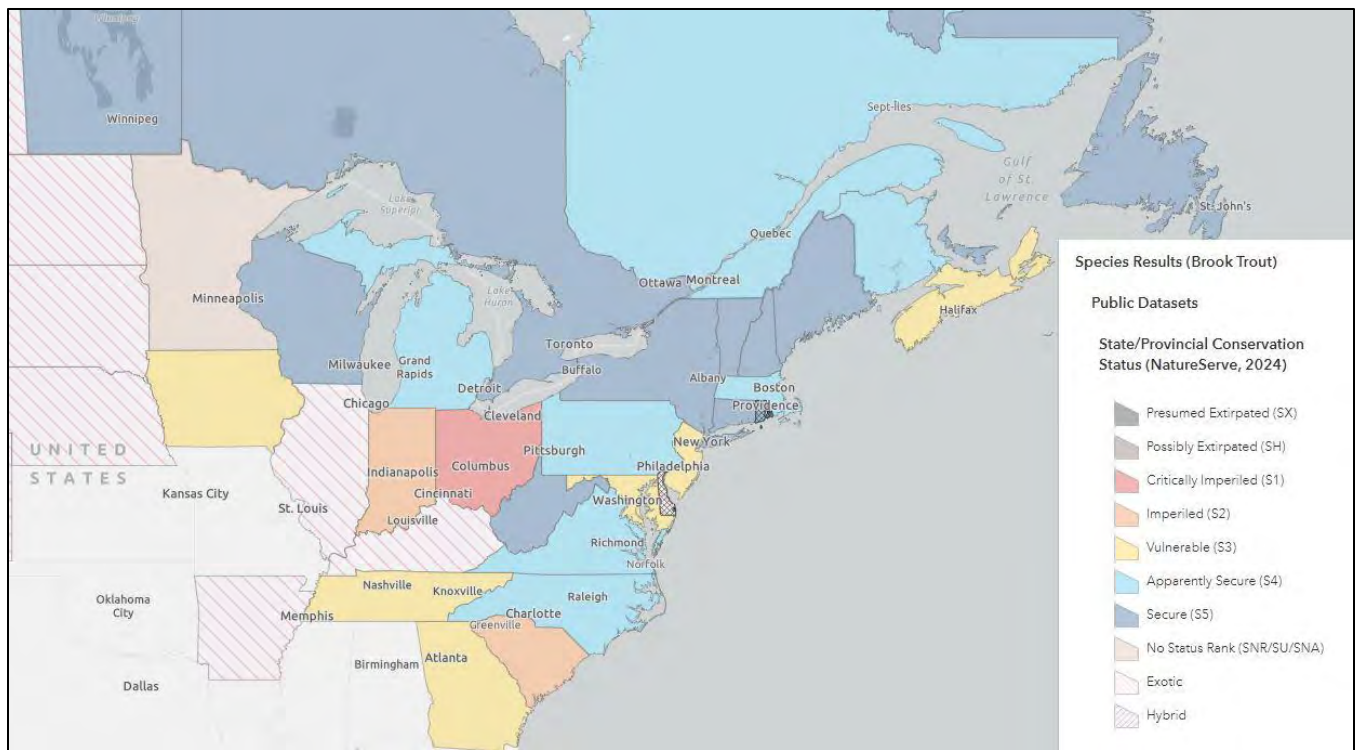


Figure 1: Brook Trout distribution and status in North America (Source: NatureServe 2024).

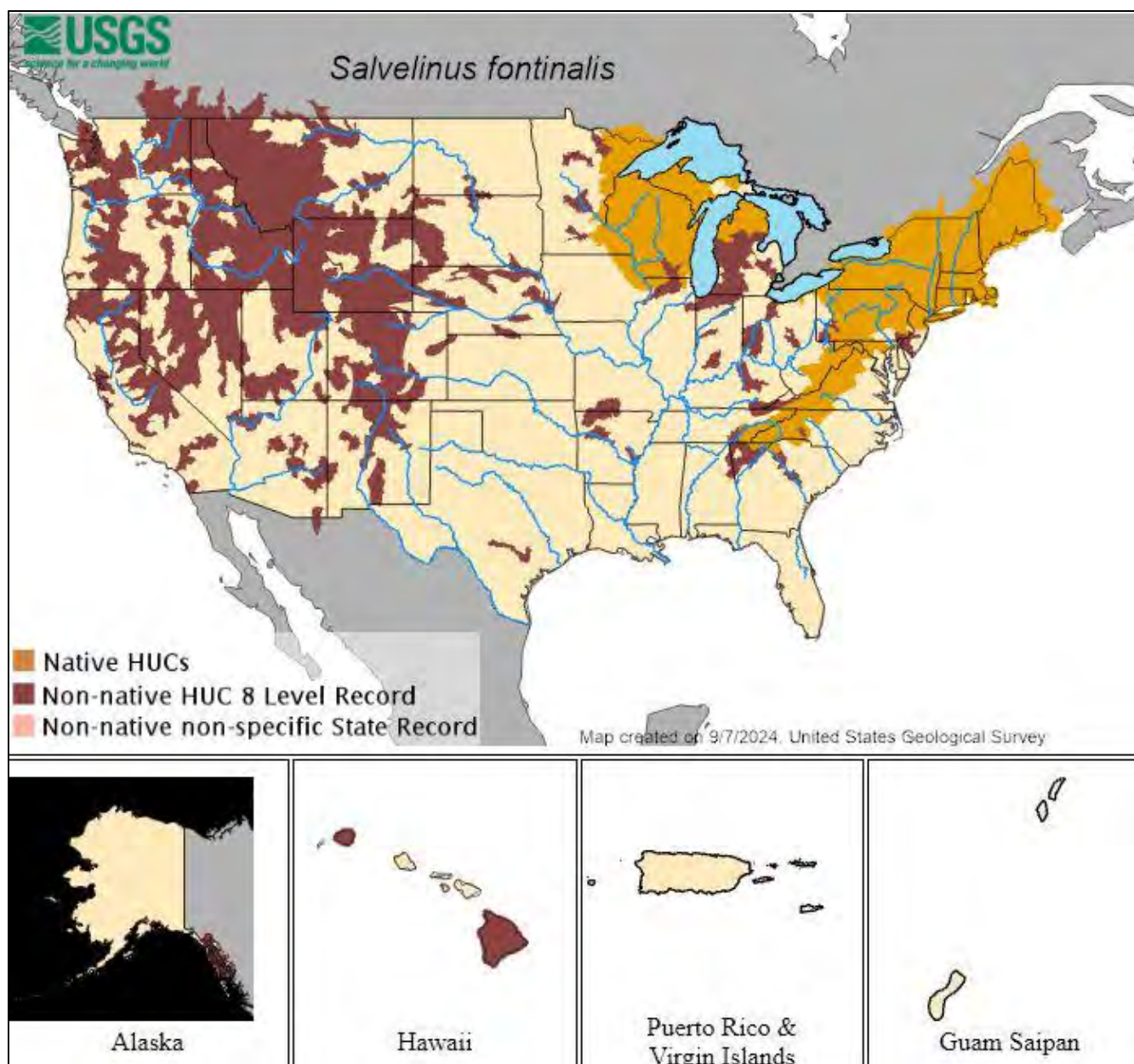


Figure 2: Brook Trout distribution in North America (Source: USGS 2024).

III. New York Rarity (provide map, numbers, and percent of state occupied)

Brook Trout *Salvelinus fontinalis*

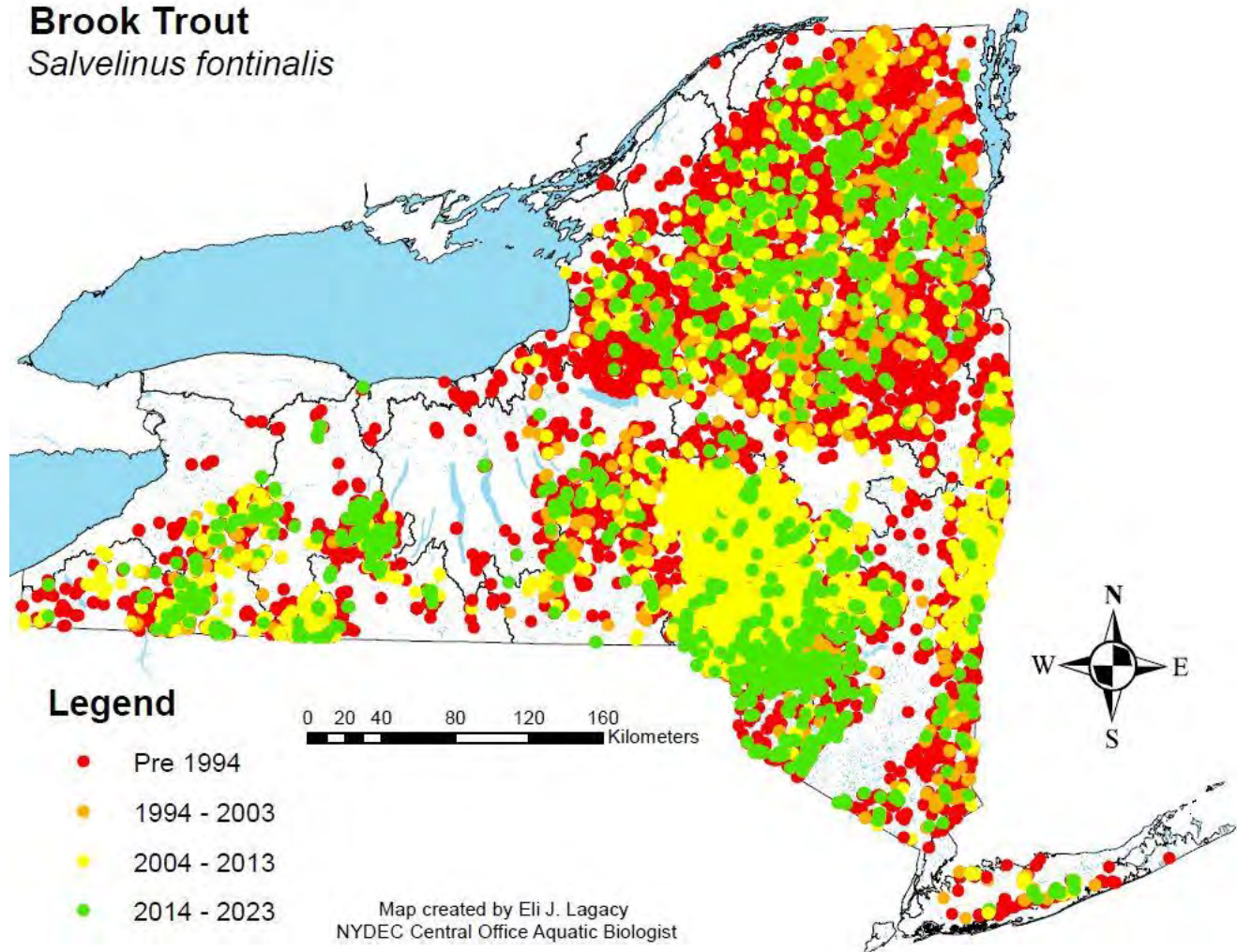


Figure 3: Records of Brook Trout in New York.

Years	# of Records	# of Waterbodies	% of State
Pre 1994	13,447	3,934	80%
1994-2003	2,487	934	70%
2004 – 2013	3,751	2,163	70%
2014 - 2023	2,621	963	70%

Table 1: Records of Brook Trout in New York.

Details of historic and current occurrence:

In New York, Brook Trout are native to all 18 major watersheds and have been introduced and subsequently managed in many other waterbodies they were originally not found (Carlson et al. 2016). In these watersheds Brook Trout were common and widespread, and generally continue to be. The only two major watersheds that have shown obvious reductions in population and distributions are Long Island and Ontario. On Long Island, upwards of 90% of the original trout

stream habitat has been lost since the 1930's, primarily due to increased urbanization (Carlson et al. 2016). Historically, there were records of brackish water populations of Brook Trout on the island as well. While small levels of natural reproduction continue to occur, the majority of Brook Trout on Long Island in recent decades are the result of stocking efforts (Carlson et al. 2016). In recent years Brook Trout have been found during surveys in streams of the central region of the island, away from the high levels of urbanization on the western end. These are: Beaver Brook, Massapequa Creek, the Connetquot River system, Tuthills Creek, Patchogue River, Swan River, Yaphank Creek, and the Carmans River.

Brook Trout were once found throughout the Ontario watershed, except in the lower elevation areas in the west. Most historical detections were in the Tug Hill region, where they are still common today, though they are now sparse elsewhere potentially due to habitat changes.

New York's Contribution to Species North American Range:

% of NA Range in New York	Classification of New York Range
100% (endemic): _____	Core: ☒ _____
76-99%: _____	Peripheral: _____
51-75%: _____	Disjunct: _____
26-50%: _____	Distance to core population: _____
1-25%: ☒ _____	<u>Within core of native distribution</u>

IV. Primary Habitat or Community Type *(from Northeast Aquatic Habitat Classification)*

a. Size/Waterbody Type: Headwaters to small rivers, lakes and ponds

b. Geology: Neutral, moderately buffered

c. Temperature: Cold

d. Gradient: Moderate-low gradient to high gradient

Habitat or Community Type Trend in New York

Declining: _____	Stable: ☒ _____	Increasing: _____	Unknown: _____
Time frame of decline/increase: _____			
Habitat Specialist	Yes: _____	No: ☒ _____	
Indicator Species	Yes: ☒ _____	No: _____	

Habitat Discussion:

The Brook Trout is found in clear, silt-free streams and small rivers, and can sometimes be found in ponds and small lakes (Smith 1985; Stauffer et al. 2016; Rice & Zimmerman 2019). There are also (albeit much less common) sea-run populations around coastal regions of the northeastern United States and Canada (Scott & Crossman 1973; Snook et al. 2015). Brook Trout are one of the most sensitive fishes in North America, and as a result are oftentimes described as an indicator species (Haxton et al. 2019). They require year-round cold, clean, water with high levels (>5 ppm) of oxygen (Werner 2004). The preferred temperature range for Brook Trout is between 11°C and 16°C, with an upper thermal tolerance of 24°C (Stauffer et al. 2016). They prefer rocky substrates and habitat with abundant cover such as large boulders, undercut banks, woody debris, and overhanging vegetation (Rice & Zimmerman 2019). Brook Trout are intolerant of excess siltation as it not only eliminates interstitial spaces for macroinvertebrates, their primary food source, but also eliminates their required spawning habitat (Werner 2004; Rice & Zimmerman 2019).

V. Species Demographics and Life History

Breeder in New York: ☒

Summer Resident: ☒

Winter Resident: ☒

Anadromous: ☐

Non-Breeder in New York: ☐

Summer Resident: ☐

Winter Resident: ☐

Catadromous: ☐

Migratory Only: ☐

Unknown: ☐

Species Demographics and Life History Discussion *(include information about species life span, reproductive longevity, reproductive capacity, age to maturity, and ability to disperse and colonize):*

Brook Trout spawn in the fall when water temperatures reach roughly 10°C, in Pennsylvania this typically occurs in mid-late September and continues through early November (Smith 1985; Stauffer et al. 2016; Holm et al. 2021). Spawning occurs over gravel bottoms in small to large streams, or along lake/pond shorelines with natural groundwater upwellings (Stauffer et al. 2016; Holm et al. 2021). Males and females will generally pair 1:1, with males establishing territories, and females building nests by pushing away gravel with her caudal fin (Stauffer et al. 2016; Holm et al. 2021). Females can construct several nests throughout the spawning season with each spawning event resulting between 40-79 eggs (Smith 1985; Stauffer et al. 2016). Brook Trout eggs are large and have been found to range between 4.1-5.2 mm in diameter (Hutchings 1991). Eggs are intolerant of high temperature with upper lethal limits of only 11.7°C, and hatch in the late winter or early spring depending on temperature and oxygen levels during incubation (Scott & Crossman 1973). Growth rates are also dependent on temperature with the fastest growth rates for alevins between 12.4-15.4°C, and reduced growth and increasing mortality as temperatures approached 19.5°C (McCormick et al. 1972). Maturity can be attained by age-1, but more often occurs at age-2 (Wydoski & Cooper 1966). Brook Trout rarely live beyond age-5 in the wild (Scott & Crossman 1973).

VI. Threats *(from NY CWCS Database or newly described)*

Brook Trout require cold, clean water, high oxygen content, substrates free of silt, and are sensitive to changes in their environment (Smith 1985; Werner 2004; Stauffer et al. 2016). Thus, modifications to waterbodies that alter these habitat conditions will be threats to their persistence. Today, the most common threats to Brook Trout habitat are land use changes and riparian clearing due to agricultural activities, lumbering, and development (Stauffer et al. 2016). The EBTJV conducted a study in 2005 identifying the key stressors throughout a portion of the Brook Trout's native range. They found that four of the top five stressors were habitat related and were as follows: "increased water temperatures (20%), agriculture (15%), urbanization (10%), one or more exotic fish species (7%), and degraded riparian habitat (7%; NJCFMP 2005)." Additionally, reduced stream connectivity from dams and culverts may also lead to a reduction in habitat connectivity and availability for Brook Trout, resulting in reduced dispersal and spawning potential (Nislow et al. 2011). Other threats to Brook Trout include biotic interactions with introduced non-native species. Brown Trout (*Salmo trutta*) and Rainbow Trout (*Onchorynchus mykiss*) are two species that have been introduced to much of the Brook Trout's native range for sport. Several

studies have shown that Brown Trout especially can outcompete Brook Trout for resources and have negative impacts on their populations and distributions (Fausch & White 1981; Hitt et al. 2016; Zorn et al 2020).

Are there regulatory mechanisms that protect the species or its habitat in New York?

Yes: ✓

No:

Unknown:

If yes, describe mechanism and whether adequate to protect species/habitat:

The Protection of Waters Program provides protection for rivers, streams, lakes, and ponds under Article 15 of the NYS Conservation Law which regulates excavation and fill in waters of New York. However, not all streams are protected, and agricultural activities are exempted from the Part 661 regulations. Additionally, New York state has implemented fishing regulations to limit the impact of sport fishing and harvest on Brook Trout populations. From October 16 – March 31, Brook Trout are to be targeted with artificial lures only and released, harvest can only occur from April 1 – October 15 and is limited to 5 fish/day and only 2 over 12.”

Describe knowledge of management/conservation actions that are needed for recovery/conservation, or to eliminate, minimize, or compensate for the identified threats:

The New York State Trout Stream Management Plan (2020) has changed the way that trout fisheries are managed throughout the state. Under the new plan, stream reaches will be managed “based on a combination of their ecological and recreational potential, with a clear distinction between wild trout and stocked trout management.” With a renewed emphasis placed on wild trout management, Brook Trout are now a large focus under the plan and self-sustaining populations along with their habitat will be high priorities within the state. Additionally, New York State implements harvest and size restrictions on Brook Trout with no harvest between October 16 – March 31. From April 1 – October 15 only five (5) Brook Trout may be harvested per day, with only two (2) above 12 inches.

Habitat management and protection will be crucial in order to maintain adequate in-stream habitat for wild Brook Trout. Sustaining and improving habitat may increase resilience to additional threats such as environmental disturbances brought upon by global climate change. One of the top threats to wild Brook Trout populations are increasing water temperatures (NJCFMP 2005). Working with local landowners and organizations such as the EBTJV and Trout Unlimited to reduce or lessen alterations that would increase water temperature will benefit wild Brook Trout populations and other cold-water dependent species within the state. Limiting the impacts of agriculture and urbanization by maintaining robust riparian buffers along streams in altered areas will help to retain natural processes Brook Trout depend on for persistence, and limit excess deposition of fine substrates. Additionally, ensuring proper stream connectivity will help to preserve adequate habitat availability. This includes assessing stream and river crossings and remediation, if necessary, along with the removal of dams that have been deemed unsafe or have little to no economic value (provided this does not increase the invasion front for a problematic species).

Conservation Actions	
Action Category	Action
1. Land/water protection	Site/area protection
2. Land/water protection	Resource & habitat protection
3. Land/water management	Site/area management
4. Land/water management	Invasive/problematic species control
5. Land/water management	Habitat & natural process restoration
6. Species Management	Harvest Management
7. Species Management	Species recovery
9. Law & Policy	Legislation
10. Education & awareness	Awareness & communications

Table 2: Recommended conservation actions for Brook Trout.

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